

Research Paper on EV Adoption and Policy Incentives

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1 Introduction

EV penetration in the automobile industry Electric vehicles (EVs) have emerged as a pivotal technology in the global transition towards sustainable transportation. Their adoption is closely linked to environmental policy frameworks, which aim to mitigate greenhouse gas emissions and reduce urban air pollution. Governments worldwide have implemented a range of incentives to accelerate EV uptake, including direct purchase subsidies, tax rebates, and exemptions from registration fees. In addition, indirect incentives such as investments in charging infrastructure, preferential access to urban zones, and public awareness campaigns have played a significant role in shaping consumer behavior. The effectiveness of these policies depends on local market conditions, technological readiness, and the broader regulatory environment. As EV technology continues to evolve, policy incentives remain a critical lever for driving adoption and achieving environmental objectives.

- Global context of policy incentives for EV adoption

- Review on three streams of literature: This study primarily relates to the following three strands of literature.

Differentiated Market Demand estimation First, this study closely relates to the mature literature on differentiated product demand estimation, in which the automobile market has been specifically focused on with a variety of directions for extensions. Following the foundational work of MacFadden (1977), the structural approach of logit discrete choice has been widely adopted for the demand-supply style problems. To allow for a more realistic assumptions and promptly deal with dimensionality problems, variants in the model such as the nested logit model, the principles-of-differentiation general extreme-value (PD-GEV) model in Bresnahan et al. (1997), and the adopted approach in this paper of random-coefficient discrete choice model in Berry et al. (1995). Following the seminal works of Bresnahan (1987), Berry et al. (1995), and Petrin (2002), I apply the standard but rigorous structural modelling style to the automobile

industry, to maintain flexible substitution patterns and facilitate counterfactual policy evaluations. Although automobile demand estimation has been thoroughly discussed in the past three decades, the focus on EV is relatively fresh within the literature. Li et al. (2017) began with a stylized model to evaluate the effectiveness of EV and infrastructure incentives in the U.S. market. They found that charging station subsidies of equal sizes are twice as effective for EV adoption than purchase incentives. Springel (2021) aimed at similar question in the Norwegian market, but implemented a structural demand-supply specification and estimated a random coefficient model under the influence of indirect network externality of charging stations, which then facilitates the testing of counterfactual policy mixes. Li (2023) considered again the U.S. automobile market and EV adoption using a BLP-style model, but introduced incompatible charging standards to the supply side and focused primarily on the impact of a uniform-standard policy for EV penetration. Remmy (2023) expanded the structural model by endogenizing supply-side decision on range of EVs and explored firms', in addition to consumers', responses to incentive schemes in Germany. Finally, Heid et al. (2024) enriched the modelling further by bringing in electricity market to form a joint-equilibrium model, endogenizing charging timing on the demand side, and further discussed the issue of pass-through in incentives.

As for studies on the Chinese automobile industry, Jia et al. (2023) studied the impact of attribute-based incentives to EV adoption,

Indirect Network Effect The interaction between EV and charging station presents a typical example of a two-sided market, in which the presence of indirect network effect would generally alter policy implications significantly (Armstrong 2006). This study specifically adds to the strand of automobile demand modelling by testing the result of indirect network effect in the Chinese market.

EV Adoption and Environmental Policy Lastly, this paper contributes to the understanding of policy impacts on EV adoption. The Chinese government has prompted EV penetration of the largest in scale and fastest in speed through strong policy supports. Although the growing literature of EV adoption and environmental policy has studied cases of various developed economies such as Norway (Springel 2021), the US (Li et al. 2017) (Li 2023), and Germany (Remmy 2024), there is yet to be a systematic study of the evolving Chinese EV market to my knowledge. EV policies have been pivotal to the net-zero target of China, where a wide-range of policies have been entering, substituting and existing in a frequent basis, allowing for many additional policy questions to be answered by evaluating the rich set of policy mix. In particular, the question on policy exit effect is rarely asked for this relatively young industry, and this paper adds to that direction, by extending the question of optimal policy mix in Springel (2021) to exploring the compensatory effects of policy substitutions.

2 Industrial Background and Policy

- **Development of Chinese EV market** The Chinese EV market has experienced a rapid expansion period over the last decade, and various incentives in the form of both central and disaggregated policies have contributed to the process. Back in 2009, the Chinese government first initiated the experimental policy initiative of "1,000 EVs in 10 cities", aiming at introducing new energy vehicles at 10 cities per year. To initialize EV adoption, the replacement over traditional ICE cars initially took place in public transports, taxi, and other government-led fields. However, EV penetration was unsuccessful during earlier stages due to immature industry of EV production, technologies and infrastructures. Within the same period, monetary incentives were also introduced on both supply and demand side in the form of cost rebates and purchase subsidies. In 2014, a uniform tax rebate policy was introduced for all purchases on BEVs and PHEVs. These continued to exist until the environmental policies gradually gained public attention in major cities. Many local environmental policies had taken the form of travel and registration restrictions on traditional vehicles while leaving EV unaffected. As technologies progressed, price effects combining with usage value rise accelerated EV penetration.

In the starting period, EV share rose explosively but with a strong reliance on intense monetary incentives. The side effects of subsidy fraud also came as a side effect. As a result, the attention gradually shifted from direct to indirect policies, among which infrastructure supply was key due to the unique two-sided market feature of EV. From 2017, policy exists on direct subsidies was complemented by construction cost rebate and operational subsidies for charging stations. In the meantime, foreign capital restrictions were lifted for automobile producers, following the entry of Tesla's super-factory in Shanghai, number of brands began to rise, pushing up the intensiveness of differentiated product competition. In addition, the 'Dual Credit Policy' published in 2017 effectively ensured a minimum share for EV from the supply side, resulted in large scale entries of traditional ICE automobile players to expand into EV production.

- Central/local government incentives: purchase and infrastructure

3 Data

The data I use is compiled from a variety of sources, including a commercial source that aggregated registry data from the Chinese Association of Automobile Manufacturers (CAAM), indicative price and range data from the Ministry of Industry and Information Technology (MIIT), charging station stock data from the China Electric Vehicle Charging Infrastructure Promotion Alliance (EVCIPA), the province-level demographic data National Bureau of Statistics (NBS), and supplementary information on a wide-range of government incentives at different levels.

The primary data set is the sales data where each entry contains the number

of sales for a particular model within a province in a given year in the range of 2019-2023, it also contains brand, fuel-type and two supplementary product characteristics of each model: mass and power. This data set is over all types of private cars, consisting ICE cars and EVs that can be further divided into BEVs and PHEVs. The data set was purchased from a commercial source that aggregated micro data of Compulsory Traffic Accident Liability Insurance (CTALI) maintained by the CAAM. It is common practice to use this measure as proxy for car purchases in the Chinese market, as CTALI registered cars are the ones actually in-use. The number of models appears to vary across years and the availability of some are not uniform across provinces. I define a market as the unit of observation at the model-year-province level, the data set consisting of ?? models, 31 provinces over a 5-years interval, gives me a total number of ?? observations. This sample size is significant compares to other studies due to the large geographic dimension and uniquely rich diversification in the Chinese automobile market, supported by its market base.

The second data set collected information on indicative price range published by the MITT, and I use the midpoint of which as the market price. It also contains the range of each model.

I also obtain the charging station stock at the province-year level from 2018-2023.

Lastly, I combine mean and standard deviation of demographic variables for each province-year.

4 Empirical Modelling

4.1 Demand Modelling

For modelling vehicle demand side, I follow the traditional setup of differentiated product demand estimation to use a discrete choice specification with random utility maximization. In particular, assume there are $m = 1, \dots, M$ markets that is defined as a province-year pair within my context of study. Within each of the defined market, the modelling is on individual purchase decisions of $i = 1, \dots, I_m$ number of potential consumers, made over $j = 1, \dots, J$ different vehicle models in the product space. The indirect utility of consumer i , $U(i)$ of consuming product j in market m is specified as a function of both observed and unobserved product characteristics:

$$U_{ijm} = \alpha(y_i - p_{jm}) + x_{jm}\beta^{ex} + \xi_{jm} + \bar{\epsilon}_{h(j)m}^i + (1 - \rho)\bar{\epsilon}_{ijm} \quad (1)$$

where p_{jm} denotes the net product prices that includes taxes and intermediate markups, y_i the income of consumer,

5 Results

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6 Policy Counterfactuals

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7 Conclusion

8 Literature Review